Anchored Quadrature Cone Tracing: Deterministic, Cache-Amortized Voxel Cone Tracing for Real-Time Global Illumination

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Fig. 1. AQCT in Khronos Sponza at 1920 × 1080 on an Apple M5, all frames from one real-time configuration (three diffuse bounces, 64 cone pairs per probe, ray-traced sun). From left to right: static camera (11.6 ms per frame), moving sphere with an exact analytic shadow (12.8 ms), sun moving at 0.1° per frame (13.2 ms), and camera dolly at 2 cm per frame (16.2 ms). Indirect light comes from persistent, world-anchored screen probes that re-trace a rotating sixteenth of a fixed direction set each frame.

Voxel cone tracing gathers diffuse indirect light from a prefiltered voxel volume at a cost set by the number of cones rather than by scene complexity. In practice its cost is dominated by per-fragment gathering, and rotated or randomized cone sets make each estimate depend on temporal filtering. We present *Anchored Quadrature Cone Tracing* (AQCT), which keeps Crassin's cone march and changes the estimator and the schedule around it. The estimator is a deterministic, world-space set of antipodal Fibonacci directions with per-normal cosine weights. Because a probe re-traced on an unchanged surface reproduces the same value, the estimate can be cached: persistent screen-space probes trace the set once, re-trace a rotating subset each frame so that any lighting change reaches every direction within a bounded number of frames, and are blended with a smooth-lighting interpolation that removes the tile-grid artifacts of naive probe interpolation. Around these sit a coverage-weighted voxelizer, a stratified sun-visibility cache with an exact analytic moving-object shadow, and an amortized lighting schedule. On an Apple M5 the system renders Sponza at 1080p in 11.6 to 16.3 ms per frame across static, camera, sun and object motion. A matched ablation shows that the deterministic set is no more *accurate* than a rotated set once cone count and aperture are equal; its benefit is that it can be cached. An accuracy gain we had previously attributed to the quadrature came from a 1.4× difference in cone aperture.

CCS Concepts: • **Computing methodologies** → **Rendering**.

Additional Key Words and Phrases: global illumination, voxel cone tracing, irradiance caching, screen-space probes, quadrature, real-time rendering

1 INTRODUCTION

Diffuse global illumination in real time is a trade between precomputation, probe grids, screen-space methods and hardware ray tracing. Voxel cone tracing (VCT) [Crassin et al. 2011] occupies a useful point in that space: one prefiltered volume answers diffuse, glossy and occlusion queries; the cost of a query depends on the number of cones, not on triangle count; and further bounces follow from re-injecting gathered light into the volume. Crassin et al. trace about five wide fixed cones for diffuse light from every fragment and report 25 to 70 frames per second on a GTX 480.

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Three costs remain in a per-fragment design. *Sampling*: a few wide cones under-resolve the hemisphere, and adding or randomizing cones exchanges bias for noise that only temporal filtering removes. *Placement*: every fragment launches its own cones, although incoming irradiance at a world position and normal varies slowly across surfaces and over time. *Scheduling*: a moving light or object forces either a full re-gather or none, with no way to spend a fixed fraction of the cost per frame and still converge to the full answer.

AQCT addresses these by changing what surrounds the cone march (Figure 2). Our contributions are:

- a **world-anchored hemisphere quadrature** for VCT, made of antipodal Fibonacci directions with normalized per-normal cosine weights, whose estimate is reproducible across frames and re-traces, together with a matched ablation that separates its effect from cone count and aperture (Sections 3.4 and 4.2);
- a **persistent screen-probe cache** built on that property: probes are traced once, refreshed by rotating direction subsets, and re-placed without visible pops (Section 3.5);
- a **smooth-lighting probe interpolation** and a drift-aware variant that remove tile-grid discontinuities and push/pull artifacts under camera motion while reducing cost (Section 3.6);
- a **sun-visibility cache** with stratified samples and an exact analytic shadow for a moving object, and a lighting **amortization schedule** validated against converged references (Section 3.7);
- an evaluation on an Apple M5 in which every number either comes from runs made for this paper or is labelled with its earlier source and device (Section 4).

We do not claim that caching irradiance is new; our probes follow irradiance caching and screen-space radiance caching. We do not claim to match production ray-traced GI: our reference is a CPU path tracer that shares geometry, BRDF and sky code with the renderer. The renderer is a standalone Rust and wgpu research system that

re-implements Filament's BRDF terms; it is not an integration into Filament or any engine.

2 RELATED WORK

Voxel cone tracing. Crassin et al. [2011] build a sparse voxel octree of 3^3 bricks storing *incoming* radiance as Gaussian lobes, inject direct light by splatting one photon per light-view pixel, and add anisotropic voxels (six directional values) at non-leaf levels so that opposing walls do not bleed into each other at coarse levels. Cones step along their axis, sample the level whose voxel size matches the cone diameter, and accumulate front to back with the opacity correction $\alpha' = 1 - (1 - \alpha)^{d'/d}$. The paper reports 22.0 ms for direct plus indirect diffuse and specular in Sponza at 512^2 with three diffuse cones and one specular cone, and states that the method does not suffer from noise or temporal discontinuities. Our volume follows this design (isotropic finest level, directional mips above it) but is dense, and it stores *outgoing* diffuse radiance only. Open-source VCT renderers such as VoxelConeTracingGI [compix [n. d.]] use dense clip levels, rasterized voxelization, fixed world-space cone directions and a minimum specular aperture; we used it as a design reference only and did not build or run it.

Irradiance caching and probes. Irradiance caching [Ward et al. 1988] stores slowly varying irradiance against a position and normal and reuses it within a validity region. Real-time systems place such caches on a world grid with ray-traced updates and visibility-aware interpolation [Majercik et al. 2019, 2021] or on the screen, as in Lumen's screen-space radiance cache [Wright et al. 2022]. Light propagation volumes [Kaplanyan and Dachsbacher 2010] propagate injected light through a coarse grid and share VCT's leak-prone discretization. Our probe tier follows the screen-space placement idea; the voxel-cone backend, the deterministic direction schedule, the interpolation and the re-placement rules are ours.

Temporal filtering and specular antialiasing. We use established components: luminance-weighted history blending [Karis 2014], variance clipping of the history neighbourhood [Salvi 2016], a Catmull–Rom history filter as surveyed by Yang et al. [2020], and specular antialiasing by roughness widening [Guy and Agopian 2018; Toksvig 2005] grounded in NDF filtering [Kaplanyan et al. 2016].

3 ANCHORED QUADRATURE CONE TRACING

3.1 Problem Statement and Overview

For a surface point $\mathbf{x}$ with shading normal $\mathbf{n}$ we want the diffuse irradiance $E(\mathbf{x}, \mathbf{n}) = \int_\Omega L_i(\mathbf{x}, \boldsymbol{\omega}) \max(0, \mathbf{n} \cdot \boldsymbol{\omega})\, \mathrm{d}\boldsymbol{\omega}$, where L_i is read from a prefiltered voxel volume by cone tracing. A per-fragment estimator evaluates E from scratch for every pixel and frame. AQCT instead (i) fixes the direction set in world space, so that the estimate at a given $(\mathbf{x}, \mathbf{n})$ is a deterministic function of the volume; (ii) evaluates it at sparse screen-space probes and keeps each probe's per-direction radiance across frames; and (iii) refreshes a rotating subset of each probe's directions per frame, so lighting changes propagate within a known number of frames. Algorithm 1 summarizes a frame.

Algorithm 1: One real-time frame (lighting interleave 2).

```
G-buffer and depth pre-pass;
if frame is even then
    inject direct light and one of 16 bounce slices into the volume;
else
    rebuild anisotropic mips of dirty 8^3 bricks; re-trace one of 16
      probe direction subsets;
foreach 16 px tile do
    if the tile has no valid probe, or its probe is occluded, drifted or
      past its staggered age then
        place a probe at the tile centre; trace all 64 pairs of the
          stable set;
trace 8 px adaptive probes for pixels that match no tile probe;
prefilter probe radiance into 8 × 8 hemispherical irradiance with the
  gather's cosine weights;
interpolate irradiance per pixel with smooth-lighting weights
  (Section 3.6);
update the sun-visibility cache and apply the analytic sphere
  shadow;
shade; temporal resolve;
```

The renderer has two configurations that we keep separate throughout. The *real-time configuration* (all 1080p results) uses one scene-wide volume of $256 \times 128 \times 256$ voxels (11.8 cm voxels for Sponza; 8.4 M cells, 10.0% occupied), screen probes, half-resolution specular and the amortized schedule of Section 3.7. The *fidelity configuration* (reference comparisons at 640×360) uses a per-pixel gather with up to 64 cone pairs and a 128-voxel-class volume. A two-level toroidal clipmap exists for larger scenes, but it cannot be combined with screen probes and is not used in any result here.

3.2 Radiance Volume and Coverage Voxelization

The volume stores an isotropic base texture (radiance premultiplied by opacity, opacity in alpha) and a six-face anisotropic mip chain. Sampling blends the base and anisotropic level 0 below LOD one and uses the directional mips above it, so narrow cones pay a single fetch near the surface. Mips are filtered in linear, alpha-premultiplied space; emission accumulates in HDR and is never clamped. Only outgoing diffuse radiance is stored, weighted by $1 -$ metallic; metal and glossy interreflections are therefore not transported.

Each triangle is covered by equal-area surface samples, about four per voxel face. Each sample splats trilinear weights, scaled by its projected area, to the eight nearest voxel centres. Opacity is $\min(1, 2c)$ for coverage c, times material opacity: sheets stay watertight and opacity varies continuously under sub-voxel motion. Sums use fixed-point integer atomics (scale 2^{20}), so the result does not depend on thread order. GPU tests check that a plane moved in 0.1-voxel steps stays opaque in every column with per-step opacity change below 0.25, that splitting a patch into 2 or 128 triangles changes the volume by less than 5% of its peak, and that an equal-area red/blue patch stays a 50/50 mixture regardless of tessellation. Initial voxelization of Sponza takes 12.9 ms on the M5 GPU.

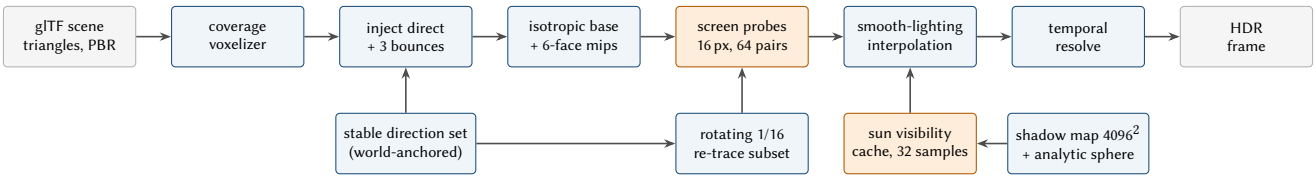


Fig. 2. Frame structure. Orange: world-anchored caches that persist across frames. One stable direction set drives both per-voxel bounce gathering during injection and per-probe gathering for shading.

3.3 Cone March

Cones are marched in steps of 0.75 of the local cone diameter, with front-to-back accumulation and Crassin's opacity correction, a transmittance cut-off of 0.9995 and at most 512 steps. An exact empty-space skip is implemented but disabled in every result reported here.

3.4 World-Anchored Quadrature

For $k \le 64$ the stable set contains, for $i = 0, \dots, k-1$,

$$\mathbf{d}_i = \left(\sqrt{1-y_i^2}\cos\phi_i,\ y_i,\ \sqrt{1-y_i^2}\sin\phi_i\right), \tag{1}$$

with $y_i = (i+\frac{1}{2})/k$ and $\phi_i = i\,\pi(3-\sqrt{5})$, together with its antipode $-\mathbf{d}_i$, giving $2k$ world-space directions of which about k lie on any hemisphere (Figure 3). Antipodal pairs share one storage slot in a probe. For a shading normal $\mathbf{n}$ the weight of a direction is

$$w(\mathbf{d}) = \frac{\max(0, \mathbf{n}\cdot\mathbf{d})}{\sum_j \max(0, \mathbf{n}\cdot\mathbf{d}_j)}, \qquad \hat{E}(\mathbf{n}) = \pi \sum_{\mathbf{d}} w(\mathbf{d})\, L_i(\mathbf{d}). \tag{2}$$

The weights sum to one. For a cosine lobe the normalized first moment is $\frac{2}{3}$; numerically $\sum w(\mathbf{d})\,\mathbf{d}\cdot\mathbf{n}$ is 0.666, 0.667 and 0.667 for $k = 16, 32, 64$ with $\mathbf{n} = \hat{\mathbf{y}}$, and 0.659, 0.662 and 0.662 for a tilted normal. Directions with $\mathbf{d}\cdot\mathbf{n}_g \le 0$ for the geometric normal $\mathbf{n}_g$ are skipped, and the march uses a fixed starting phase. Each cone's half-angle comes from equal-solid-angle hemisphere patches, $\cos\theta = 1 - 1/k$, times an aperture scale (0.125 in the real-time configuration).

What determinism buys. The estimate at a surface point is a deterministic function of the volume. This gives three properties that a per-pixel rotated set lacks: a static scene produces a static estimate without temporal accumulation; a probe re-placed on an unchanged surface reproduces its predecessor's value, so re-placement does not pop; and a probe's per-pair radiances can be refreshed in disjoint subsets whose union is exactly the full estimate. The cache below relies on all three. Whether the set is also *more accurate* is a separate question, answered in Section 4.2.

3.5 Screen-Probe Irradiance Cache

One probe per 16 px tile sits at the tile centre on a visible surface and traces all 64 antipodal pairs from the origin a per-pixel gather would use. Radiance is stored per pair in RGB9E5 and prefiltered with the gather's own cosine weights into an 8×8 hemispherical irradiance map (one workgroup per probe; the 64 pair radiances are decoded once into workgroup memory). A kept probe re-traces a rotating sixteenth of its pairs per frame; with lighting interleave

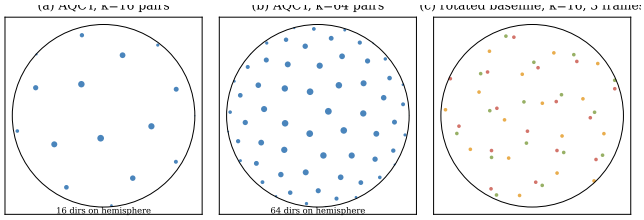


Fig. 3. Direction sets on the tangent plane for $\mathbf{n} = \hat{\mathbf{y}}$; marker area is the cosine weight. (a,b) Stable set, visible hemisphere at $k = 16$ and 64. (c) Schematic of the rotated baseline: a tangent-space Fibonacci disk with equal weights, rotated independently per pixel and per sample (three rotations shown).

two, any change reaches every pair within 32 frames. A probe is reused while its surface matches the tile within position and normal bounds. It is re-placed and fully re-traced when it becomes occluded, drifts more than about 1.5 tiles, or exceeds a staggered maximum age of 32 to 64 frames; a probe that has not left its placement pixel is exempt from age-based re-placement. Pixels that match no tile probe use an 8 px adaptive probe.

Settling and object awareness. When the camera is still, a rotating quarter of the probes that sit more than 2 px from their placement pixel is re-placed each frame, so all are re-centred within four frames, and a re-centred probe on the same surface cross-fades from its predecessor over eight frames. Kept probes within eight radii of the moving sphere's current or previous position re-trace their rotating subset plus every pair whose cone meets the sphere.

3.6 Smooth-Lighting and Drift-Aware Interpolation

Visible "blocky" indirect shadows under motion turned out to be exact 16 px screen-aligned squares. The original interpolation gave each probe a radial tent reaching two tiles but read only a 3×3 neighbourhood, so a column of probes with non-zero weight dropped out whenever a pixel crossed a tile edge. We replace it with a smooth-lighting scheme of the kind used in voxel games: each pixel blends the probes of the 2×2 tiles around it with bilinear weights on the tile-centre lattice, rejects probes on other surfaces and renormalizes the rest. Where the accepted weight falls below 0.25, the result blends continuously into a plain neighbourhood average, which covers silhouettes. The weights depend only on pixel position, so they are continuous across tiles, and only probes with non-zero weight are fetched (at most four in the common case instead of nine), which makes the pass faster as well as smoother (Figure 4).



Fig. 4. Probe interpolation during a camera dolly (M5, 1080p, frame 48; 240^2 crop enlarged 2×). Left: 3 × 3 radial-tent interpolation. Middle: smooth-lighting interpolation. Right: absolute difference ×8, showing the 16 px tile structure that tent interpolation leaves in the indirect light.

Probes are reused while they drift within about 1.5 tiles, which keeps the birth rate low in motion, but their irradiance was measured at the reprojected position. Weighting them at the tile centre drew shading up to a tile away from where it was measured: outward on a dolly forward, inward on a dolly back. The bilinear tents are therefore centred on each probe's actual screen position, and the neighbourhood grows from 3×3 to 5×5 only while the camera moves; a workgroup-shared bitmask pre-tests the outer ring.

3.7 Sun Visibility and Lighting Amortization

Cone-traced sun shadows are not competitive (Section 5), so the real-time configuration uses a software triangle-BVH ray tracer for sun visibility and caches 32 samples per pixel. The sun is a disc of 2.5° half-angle, about ten times the real sun; the reference uses the same disc. Four changes made the cache stable. History lookup measures its acceptance radius along the surface, dividing by the view–surface cosine, so grazing floors keep their history under sub-pixel jitter. The 32 samples are the points of a golden-angle (Vogel) spiral over the disc with a fixed per-pixel rotation. During camera motion only penumbral pixels receive a refresh ray, at weight 0.125. The moving sphere is applied analytically each frame as the overlap of the sun disc and the sphere's angular disc, which is exact and invalidates no cached sample.

The schedule interleaves lighting work over two frames (injection and one of 16 bounce slices on one; mips of dirty 8^3-voxel bricks and one of 16 probe re-trace subsets on the next) and renders the shadow map every frame. While the sun moves slowly (below about 0.08° per frame), a rotating quarter of pixels blends in one fresh ray per frame; faster motion re-samples every pixel; when the sun stops the cache re-converges exactly.

3.8 Temporal Resolve and Specular

The resolve uses jitter-free reprojection with the closest-surface velocity in a 3×3 neighbourhood, rejects history only when the history surface matches no current neighbour, clips history to a YCoCg variance box of 1.5σ [Salvi 2016], weights by luminance [Karis 2014] and filters history with a nine-tap Catmull–Rom kernel. Specular cones are widened by half the ray-differential footprint, start at the pixel's world footprint, and never fall below a minimum aperture; Filament's geometric specular antialiasing and Toksvig filtering widen roughness.

Table 1. Real-time frame cost on the Apple M5 at 1080p (mean of three runs of 96 frames). Stage times are GPU timestamps averaged over all frames, including frames on which an interleaved stage does not run; because passes overlap in the pipelined frame, their sum exceeds throughput.

Stage (ms)	Static	Camera	Slow sun	Sphere	Fast sun
G-buffer and depth	2.73	2.48	3.00	2.92	3.45
Lighting (inject, bounce, mips)	2.50	2.42	2.56	2.63	2.63
Probe trace (tile, adaptive)	1.30	5.21	1.29	1.60	1.31
Probe filter, interpolation	1.99	2.41	1.99	2.23	2.01
Sun (shadow map, cache)	2.00	2.12	3.20	2.21	5.37
Specular, reconstruction	2.47	2.46	2.47	2.50	2.49
Temporal resolve	1.37	1.42	1.37	1.38	1.39
Throughput	**11.58**	**16.19**	**13.18**	**12.78**	**16.33**
Frames per second	86	62	76	78	61

4 RESULTS

Setup. All new measurements were made for this paper on an Apple M5 MacBook Pro (10-core GPU, Metal 4, 24 GB unified memory, macOS 26.7.1) with a release build of the renderer (Rust 1.94.1, wgpu). The scene is Khronos Sponza [Khronos Group [n. d.]] (262,267 triangles, 26 materials, 1024^2 textures). Real-time cases use the demo view at 1920×1080; timings are pipelined throughput over 96 frames after 40 warm-up frames, repeated three times. Fidelity comparisons use the `nave` camera at 640×360 against a 512-spp reference from our CPU path tracer [Pharr et al. 2023]. We report linear-RGB RMSE, PSNR and SSIM [Wang et al. 2004] on $1 - e^{-4Y}$ tone-mapped luminance, and *geometry bias*, the mean luminance error over a material-independent geometry mask. Our scoring script reproduces the scores of an earlier study from its stored images to within 0.03 points of bias and 0.001 of SSIM.

4.1 Performance

Table 1 and Figure 6 report frame times. On the M5 every case, including a sun dragged at 1° per frame, runs faster than 16.7 ms (60 frames per second); the spread between repeats is at most 0.09 ms. Camera motion is the most expensive case because about 3 ms go to fully tracing newly born tile probes and 1.5 ms to adaptive probes. A fast sun costs 3.6 ms of per-pixel ray refresh. For context, an earlier build (October 14) measured on an Apple M1 ran the same five cases in 28.8, 38.2, 39.9, 37.2 and 48.7 ms; those numbers come from a different build and are not a measurement of the configuration in this paper.

4.2 Is the Quadrature More Accurate?

Synthetic test. We first isolate the direction set from the renderer (Table 2). Radiance is an analytic function, point-sampled with no cone footprint; the target $E(\mathbf{n})$ is integrated on a 4×10^5-point sphere lattice for 400 random normals. The stable set with k pairs is compared with a rotated tangent-space Fibonacci disk of k equal-weight, cosine-distributed directions, rotated independently per normal, as one frame and as a 16-frame average. For smooth sky radiance the stable set is 1.7× more accurate than one rotated frame at equal count, and it is exactly repeatable. For broad patches it is 1.2× worse at $k = 16$ and 1.1× better at $k = 64$. For a compact 3° lobe



Fig. 5. Decomposition of the static frame of Figure 1 (M5, 1080p). From left to right: albedo, direct light only, indirect diffuse from the probe cache, final image.

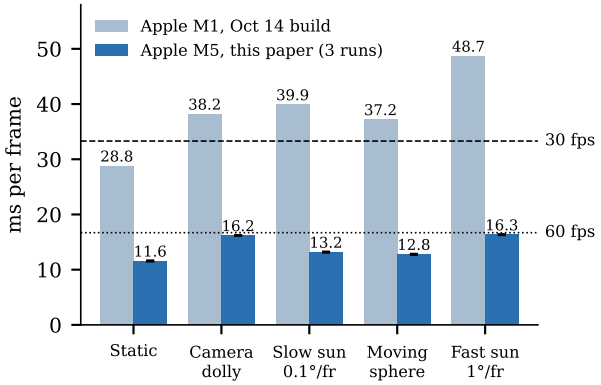


Fig. 6. Frame time per motion case. Apple M5: this paper's configuration, three runs (error bars show the range). Apple M1: an earlier (October 14) build, shown for context only.

Table 2. Synthetic test: relative RMS irradiance error (lower is better) of point-sampled analytic radiance at equal directions per hemisphere. Rot. 1: one rotated frame. Rot. 16: average of 16 frames. Single seed, 400 normals.

Radiance	k	Stable	Rot. 1	Rot. 16
Smooth sky	16	0.0077	0.0129	0.0034
	64	0.0020	0.0034	0.0009
Three broad patches	16	0.1988	0.1706	0.0459
	64	0.0257	0.0293	0.0076
Lobe, 10°	16	0.6431	0.5529	0.1284
	64	0.1348	0.1041	0.0247
Lobe, 3°	16	0.6377	2.4062	0.7550
	64	0.6365	1.3030	0.3329

it aliases: whether it hits the lobe is fixed per normal, so its error is a persistent bias rather than noise. Sixteen rotated frames beat both single-frame estimates in every case but one (the 3° lobe at $k = 16$, which every estimate mostly misses), because they use 16 times as many directions. These results already suggest that determinism, not accuracy, is the property to rely on, and they are one reason the sun is handled by a separate estimator.

In the renderer. We then run the fidelity configuration with each pattern at $k = 16, 32, 64$ (Table 3, Figure 7). At its default aperture the stable set is worse than the rotated set at every count and sample rate, with 3.9 points more negative geometry bias. The cause is in the code: the rotated set's half-angle has $\tan\theta = 1/\sqrt{k}$, while the stable set's equal-solid-angle rule gives a tangent 1.43 to 1.49 times

Table 3. Fidelity configuration on the M5 (640 × 360, three bounces, ray-traced sun) against the 512-spp reference. "Matched" scales the stable set's aperture to the rotated set's.

Set	k	Bias	RMSE	PSNR	SSIM
1 sample per pixel					
Rotated	16	−30.68%	0.02356	25.70	0.7682
	32	−26.79%	0.02211	26.51	0.7991
	64	−22.72%	0.02107	27.17	0.8277
Stable	16	−34.54%	0.02513	24.77	0.7701
	32	−30.70%	0.02377	25.41	0.8004
	64	−26.61%	0.02218	26.43	0.8166
Stable, matched	16	−27.71%	0.02359	25.42	0.7825
	32	−26.68%	0.02262	25.99	0.8163
	64	−22.79%	0.02115	27.07	0.8290
16 samples per pixel					
Rotated	16	−30.65%	0.02213	26.71	0.8204
	32	−26.75%	0.02085	27.53	0.8396
	64	−22.75%	0.01988	28.20	0.8570
Stable	16	−34.53%	0.02423	25.29	0.7915
	32	−30.70%	0.02280	26.03	0.8223
	64	−26.61%	0.02112	27.22	0.8385
Stable, matched	16	−27.69%	0.02256	26.03	0.8043
	32	−26.68%	0.02156	26.72	0.8390
	64	−22.80%	0.02002	28.01	0.8515

larger, and wider cones read more prefiltered occlusion. With the stable aperture scaled to match, bias agrees with the rotated set to within 0.15 point at $k = 32$ and 64, and at $k = 64$ RMSE, PSNR and SSIM agree to within 1%, 0.2 dB and 0.006 at one sample or sixteen. Sub-pixel supersampling, during which the rotated set re-rotates for every sample and the stable set does not, improves both sets' RMSE by the same 5%; most of that gain is geometric antialiasing rather than extra directions. A wider rotated set of five or six cones, closer to Crassin's choice, is reported in Table 4.

We conclude that at equal count and aperture the deterministic set is as accurate as a rotated one, not more. An earlier M1 study credited a 6.1% RMSE improvement to the quadrature; it compared stable 64 with rotated 16, which confounds cone count, and the aperture difference identified here confounds it further. The quadrature's contribution is the determinism that makes the cache of Section 3.5 possible. A practical consequence is that the stable aperture rule should be narrowed by about 0.7× in the fidelity configuration; the real-time configuration already uses half the fidelity aperture scale.

With our volume, few wide cones read far more prefiltered occlusion than many narrow ones and are 21 points darker; the result

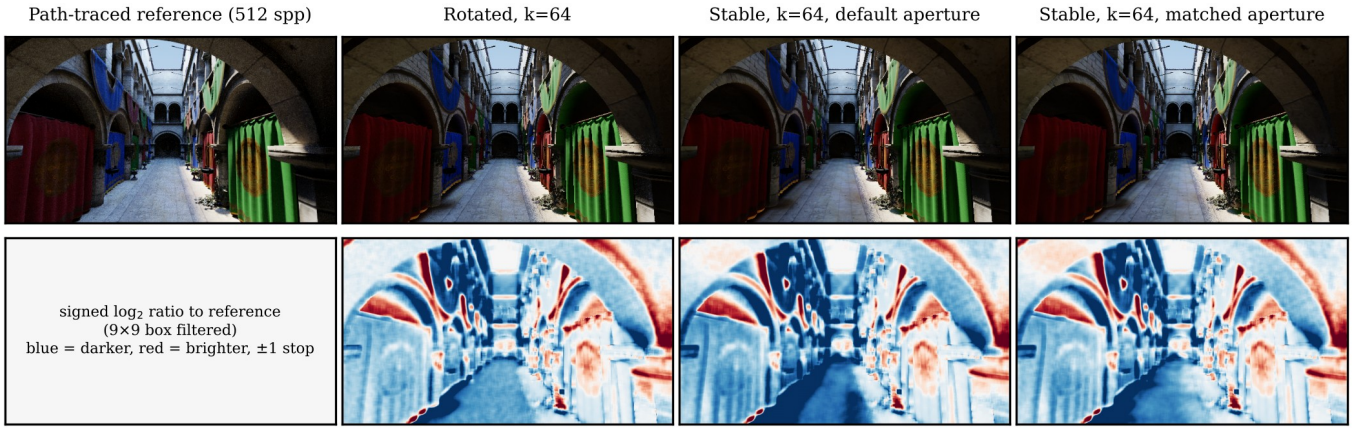


Fig. 7. Fidelity configuration at $k = 64$, one sample per pixel (M5). Top: tone-mapped images. Bottom: signed $\log_2$ luminance ratio to the reference after a 9×9 box filter, ±1 stop. At its default aperture the stable set is darker on the floor and arches (blue); with the aperture matched to the rotated set, the two error maps are nearly identical.

Table 4. A Crassin-style cone budget inside our renderer: five or six rotated cones of 30° half-angle, same fidelity configuration as Table 3. This uses our volume, voxelization and offsets and is not a reproduction of Crassin et al.'s system.

Set	Bias	RMSE	PSNR	SSIM
5 wide cones, 1 spp	−43.62%	0.03284	22.34	0.6998
6 wide cones, 1 spp	−43.57%	0.03284	22.34	0.7004
6 wide cones, 16 spp	−43.54%	0.03224	22.63	0.7208
Rotated, $k = 64$, 1 spp	−22.72%	0.02107	27.17	0.8277
Stable, $k = 64$, matched, 1 spp	−22.79%	0.02115	27.07	0.8290

reinforces that aperture, not the direction pattern, is the main driver of bias in this renderer.

4.3 Temporal Behaviour of the Two Sets

Determinism matters most over time. We ran the real-time configuration with probes disabled, so that every pixel traces its own 64-pair gather, with a static camera, and compared consecutive final frames (64 and 65) after the temporal resolve. With the stable set, the mean absolute frame-to-frame change in display luminance is 4.1×10^{-4}; with the rotated set, which re-rotates every frame, it is 7.0×10^{-4} (1.7×), and the share of pixels changing by more than 0.01 rises from 0.36% to 0.40%. The temporal resolve hides most of the rotated set's noise in a static view, so the difference in visible flicker is small. The larger difference is cost: in this code path the per-pixel gather took 0.56 s per frame with the stable set and 8.3 s with the rotated set, whose per-pixel rotations and jittered march starts destroy texture-cache locality. Both are two orders of magnitude above the probe cache, whose trace, filter and interpolation passes cost 3.3 ms in the static case (Table 1).

4.4 Probe Cache and Interpolation

The measurements in this subsection come from the October 16 to 18 development studies on the same M5 and are reported against their original references. Against a per-pixel 64-cone gather, smooth-lighting interpolation raised indirect-diffuse PSNR from 36.29 to 36.96 dB with a static camera, from 35.38 to 36.98 dB with the moving sphere and from 35.73 to 36.51 dB with a moving sun, and left the camera dolly unchanged (34.42 to 34.32 dB). The ratio of indirect-light gradient across tile edges to gradient elsewhere fell from 1.05 to 1.007, against 0.994 for the per-pixel gather itself; on the final display images of Figure 4 the same ratio is 1.019 versus 1.011. Centre placement then traded 0.2 to 0.6 dB of gather fidelity for stability. It reduced the share of flickering pixels (per-pixel temporal standard deviation above 0.01) from 5.36% to 2.08% with a static camera and paused sphere, from 8.07% to 5.24% with the sphere moving, and from 3.60% to 1.77% with a creeping camera in the gallery view. Drift-aware interpolation improved frame-47 PSNR against a settled reference from 34.82 to 35.09 dB (dolly forward), 35.60 to 36.02 dB (dolly back) and 36.14 to 37.33 dB (strafe), with mean luminance error below 0.6% in all three. After the camera stops, the distance to the settled frame 25 frames later was 1.4×10^{-3} in display luminance, against 5.6×10^{-3} before the settling change.

4.5 Sun Visibility

With cone-traced shadows, mean visibility in pixels that the reference sees as sunlit was 0.33, and 82% of them were below 0.5; no combination of cone offset and aperture gave both few false shadows and little leakage (at 1.0° with 1 or 2 voxel offsets: 34% false shadow, 10.5% leak). With the ray-traced sun, direct light matches the reference within +0.7% to +2.2%, and the stratified cache raised direct-light PSNR against a converged reference from 36.6 to 43.6 dB in the gallery view and from 34.8 to 39.0 dB in the nave.

4.6 Rejected Alternatives

Several ideas were measured and rejected: a wide 8-cone injection set (faster, but SSIM fell from 0.779 to 0.760); a probe maximum age of 128 frames (−1.1 ms, −1.35 dB static); 32 px tiles (−1.3 to −2 ms, −4.7 dB static); refreshing the sun cache every eighth frame (−1.8 ms, −1.9 dB); hashed per-tile probe placement (0.8 dB worse than centre

placement); a 25-entry loop over the 5×5 neighbourhood (+1.5 ms, replaced by the bitmask at +0.2 ms); and occluder-first BVH rays (exact, but 11.0 to 14.5 ms on the M1).

5 LIMITATIONS

Representation. A voxel holds one radiance and opacity value, so opposing surfaces in one cell bleed into each other; anisotropic mips help at coarse levels only. Coverage voxelization reduced leaks (the darkest third of pixels went from 77% too bright to 15%) but thickened sheets, and mid-tone indirect light remains 19 to 28% too dark in every configuration measured; the best geometry bias in Table 3 is still −23%.

Transport. Only outgoing diffuse radiance is stored, so there is no metal or glossy interreflection. Geometry is static apart from the camera and one analytic sphere; general dynamic objects would need dirty-region voxelization.

Shadows. Cone-traced sun visibility is not usable (Section 3.7); the ray-traced sun used in all results is not a voxel solution.

Latency. The cache is eventually correct: a lighting change reaches every direction of a kept probe within 32 frames.

Scale. The single scene-wide volume ties voxel size to scene extent. Larger scenes need the clipmap, whose re-centring is synchronous and whole-level (about 21 s at 256^3 in an M1 test) and which cannot yet be combined with screen probes.

Validation. All results use one scene and one resolution. The reference shares code with the renderer and was rendered by an earlier build of the path tracer; a second, independent renderer is needed. With a static camera, 2.08% of pixels still flicker, for reasons we have not isolated.

6 CONCLUSION

AQCT keeps Crassin's cone march and replaces the per-fragment rotated estimator with a world-anchored quadrature evaluated at persistent screen probes that refresh in rotating subsets. On an Apple M5 it renders three-bounce voxel GI in Sponza at 1080p in 11.6 to 16.3 ms per frame under camera, sun and object motion. Our matched ablation corrects our own earlier claim: at equal cone count and aperture, a deterministic direction set is not more accurate than a rotated one. What it provides is reproducibility, and reproducibility is what lets the cache trace once, refresh incrementally and re-place probes without visible pops.

ACKNOWLEDGMENTS

Special thanks to Alannis Davis, Dr. Andrew Hogue and Jakob Anderson.

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